Multiplying terms with indices

1 Write the following expressions in simplest index form.

a $2^2 \times 2^2$

b $3^4 \times 3^{10}$

c $5^4 \times 5^3$

d $6^6 \times 6^7$

e $11^4 \times 3^4$

f $17^{44} \times 17^{48}$

g $30^{72} \times 52^{72}$

h $4 \times 5^6 \times 5^7$

i $\left(3^7\right)^5$

j $\left(83^{19}\right)^{40}$

k $2^2 \times 2^3 \times 2^4$

l $2^6 \times 2^8 \times 2^7 \times 2^5$

m $-9 \times 2^2 \times 2^6$

n $9 \times (-10)^4 \times 10^8$

o $\dfrac{5}{3} \times 4^6 \times 4^2$

p $\dfrac{2}{3} \times 7^8 \times 7^9$

2 Find the product of the following expressions.

a $3^4 \times 3^3$

b $9^2 \times 9^5$

c $2^4 \times 7^4$

d $5^5 \times 3^3$

e $\left(6^2\right)^3$

f $\left(3^5\right)^2$

g $6 \times 5^2 \times 5^3$

h $\dfrac{3}{4} \times 4^4 \times 7^3$

3 Find the value of the blank that will make the following equations true.

a $3^5 \times 3^{\square} = 3^7$

b $5^8 \times 5^{\square} = 5^{12}$

c $9^{\square} \times 9^6 = 9^9$

d $31^{\square} \times 31^{37} = 31^{56}$

e $3^{11} \times (\square)^{11} = 21^{11}$

f $(\square)^{30} \times 90^{30} = 3780^{30}$

g $\left(11^{11}\right)^{\square} = 11^{33}$

h $\left(37^{\square}\right)^{14} = 37^{700}$

Dividing terms with indices

4 Write the following expressions in simplest index form.

a $3^8 \div 3^6$

b $5^{10} \div 5^5$

c $6^{13} \div 6^6$

d $10^8 \div 10^6$

e $15^{33} \div 15^{27}$

f $21^8 \div 3^8$

g $37^{30} \div 37^{18}$

h $650^{69} \div 50^{69}$

i $\dfrac{10^{10}}{10^3}$

j $\dfrac{18^{32}}{18^{23}}$

k $\dfrac{(-7)^{11}}{(-7)^5}$

l $\dfrac{(-10)^{12}}{(-10)^4}$

m $\dfrac{-7^8}{7^2}$

n $\dfrac{16^{32}}{-16^{24}}$

o $\dfrac{5 \times 4^8}{4^2}$

p $\dfrac{4^{10}}{5 \times 4^3}$

5 Find the quotient of the following express[illegible]

a $3^9 \div 3^4$

b $4^4 \div 2^4$

c $4^7 \div 4^4$

d $4^8 \div 4^5$

e $\dfrac{2^{12}}{2^8}$

f $\dfrac{3^6}{3^3}$

g $\dfrac{5 \times 4^5}{4^3}$

h $\dfrac{6 \times 5^8}{5^6}$

6 Find the value of the blank that will make the following equations true.

a $5^{\square} \div 5^5 = 5^7$

b $6^9 \div 6^{\square} = 6^7$

c $11^{10} \div 11^{\square} = 11^4$

d $31^{\square} \div 31^{21} = 31^{13}$

e $14^9 \div (\square)^9 = 7^9$

f $(\square)^{66} \div 30^{66} = 19^{66}$

The zero index

7 Evaluate the following:

a 8^0

b 21^0

c 741^0

d $(-983)^0$

e $\left(\dfrac{-5}{8}\right)^0$

f $\left(\dfrac{7}{9}\right)^0$

g $(2 \times 13)^0$

h $4\,(2 \times 13)^0$

i $65^0 + 81^0$

j $89^0 - (-95)^0$

k $72^0 + 2^3$

l $7^0 + (8 \times 7)^0$

8 Find the value of the blank that will make the following equations true.

a $3^{\square} = 1$

b $6^{10} \div 6^{10} = 6^{\square}$

c $\dfrac{8^{\square}}{8^9} = 8^0$

d $\dfrac{9^8}{9^{\square}} = 9^0$